

ADVANCED UNIT

# Advanced Water Safety

A six-lesson unit on the data, physiology, social determinants and prevention strategies behind Victorian drowning — and how to argue for change.



# Six lessons, one question

VCHPEP143 · 144 · 145 · 148 · VCHPEM162

Why does Victoria still drown — despite knowing how to prevent it? Each lesson sharpens one part of the answer.

LESSON 1

01

Advanced data analysis — Victorian & global drowning trends

VCHPEP143 · VCHPEP144

LESSON 2

02

Physiology of drowning & CPR mechanics

VCHPEP144 · VCHPEM162

LESSON 3

03

Social determinants of drowning risk

VCHPEP143 · VCHPEP145

LESSON 4

04

Prevention strategies & international comparison

VCHPEP144 · VCHPEP148

LESSON 5

05

Diving, spinal injuries & high-risk teenage behaviour

VCHPEP143 · 144 · 147

LESSON 6 · ASSESSED

06

Assessment task — planning, drafting & peer review

VCHPEP145 · VCHPEP148

PREREQUISITE · BUILT ON YEAR 7–8

# You already know the rules. Now we ask why they aren't enough.

YEAR 7–8 · YOU CAN ALREADY

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- Read the six Australian beach flags
- Recall the five rip-current survival steps
- Execute DRS ABCD in order
- Apply the five LSV SwimSafe rules
- State the headline Victorian drowning numbers

YEAR 9–10 · WE NOW ASK

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- How do you **calculate** a drowning rate and read a trend?
- What **physiologically** happens in the four minutes after submersion?
- Which **structural factors** decide who drowns?
- Which prevention strategies are **backed by evidence** — and which aren't?
- What would **you** recommend to a Minister?

# 01

## **Advanced data analysis — Victorian & global drowning trends**

VCHPEP143

VCHPEP144

CLASSROOM · INDIVIDUAL + PAIRS

“

**Why does Victoria still drown — despite decades of awareness campaigns, training, and warning signs — and what would actually move the numbers?**

# The worst Victorian drowning year in over a decade

FATAL DROWNINGS

54

Lives lost in Victorian waters — the highest annual toll in over ten years.

NON-FATAL DROWNINGS

+60%

Above the 10-year average. The near-misses tell us the system is under more strain than the death toll alone shows.

CALD COMMUNITIES

~40%

Of all deaths — the highest proportion ever recorded in Victoria.

AT PATROLLED SITES

0

Zero of the 54 deaths occurred between the flags at a patrolled location.

HIGHEST PROPORTION EVER RECORDED

# 40%

of Victorian drowning deaths in 2023–24 were people from **Culturally & Linguistically Diverse** communities — despite CALD Victorians making up roughly half that share of the state's population.

This is **over-representation**. Lesson 3 will ask **why** — and what a school can do about it.

# Victoria · Australia · the world — per 100,000 people

VICTORIA · 2023–24

$54 \div 6,900,000 \times 100,000$

0.78

per 100,000 people · LSV 2023–24

AUSTRALIA · ANNUAL AVG

RLSSA national data

1.2

per 100,000 people · RLSSA

GLOBAL · HIGH-INCOME

WHO benchmark

~0.8

per 100,000 people · WHO

**In your workbook:** show your working. If Victoria matched the 0.8 high-income benchmark, that's about **55** → **55** deaths — but the 60% jump in non-fatal incidents is where the real warning sits. (See Activity 1.1c.)

# Drowning is one of the world's most unequal causes of death

GLOBAL DEATHS / YEAR

~320k

An estimated 320,000 people drown each year worldwide (WHO).

IN LOW & MIDDLE-INCOME

90%

Of global drowning deaths happen in low- and middle-income countries.

GLOBAL · MALE VICTIMS

~80%

Of global drowning deaths are male — a pattern repeated in Victorian data.

HIGHEST-RISK AGE

<5

Children under five are the highest-risk age group globally.

# In one sentence — what is the single most important data insight from today, and why does it matter?

Workbook open to Section 1. Notes closed. **Insight** — not just a number. The most powerful answers will name a number **and** what it implies.

# 02

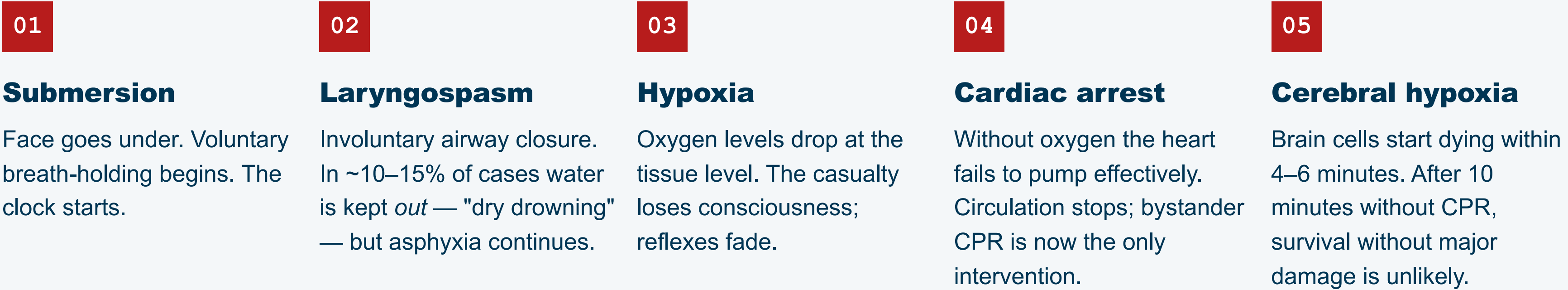
## Physiology of drowning & CPR mechanics

VCHPEP144

VCHPEM162

INDIVIDUAL

# Five stages — from submersion to brain damage



MINUTES

4

**The window in which bystander CPR makes the biggest difference between full recovery and lasting brain damage.**

Chest compressions deliver roughly **25–30%** of normal cardiac output — enough oxygenated blood to keep the brain alive until the heart can restart.

Starting at **4 minutes** produces dramatically better outcomes than starting at **8**.

# What chest compressions actually do

30

COMPRESSIONS

+

2

BREATHS

## WHY IT WORKS

Compressions create **~25–30% of normal cardiac output**. That's enough oxygenated blood to keep the brain viable until the heart restarts — spontaneously, or with a defibrillator.

Rate: **100–120 beats per minute**. Don't stop until paramedics take over or the casualty is breathing.

# Wet · dry · secondary — and the 1-10-1 cold-water rule

## A · WET DROWNING

### Water enters the lungs

The most common pattern. Water in the lungs disrupts gas exchange and triggers cardiac arrest.

## B · DRY DROWNING

### Laryngospasm seals the airway

In ~10–15% of cases, water never reaches the lungs — but the casualty still dies of asphyxia.

## C · SECONDARY DROWNING

### Symptoms hours later

**Always seek medical review** after any submersion — coughing, fatigue or chest pain can appear hours after the event.

## COLD WATER · 1-10-1

### Three survival windows

1 MIN control breathing ·  
10 MIN useful swimming ·  
1 HR before hypothermia incapacitates.

# Explain the 4-minute window to someone who has never heard of it.

Plain language. **No medical jargon.** If a Year 5 student can repeat your explanation back to you, you've nailed it.

# 03

## Social determinants of drowning risk

VCHPEP143



PAIRS · JIGSAW

# "They made a bad choice" isn't the whole story

## INDIVIDUAL EXPLANATION

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### Personal behaviour caused the outcome.

"He shouldn't have been swimming after drinking."

"They should have read the sign."

"Why did she dive in there?"

True — but incomplete. It can't explain a 40% over-representation or an 80% male skew.

## STRUCTURAL EXPLANATION

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### Systems shaped who was there, and how.

Cost of lessons. Patrol coverage. Cultural water-fluency.

Masculinity norms. Where the unpatrolled rivers are. Who has a backyard pool.

Schools can change **some** of these. That's where prevention starts.

# Six factors that shape who drowns

01 · GENDER NORMS

"Don't back down from a dare."  
Masculinity expectations are a documented risk factor for young male drowning — and explain part of the 80% male skew.

02 · CULTURAL BACKGROUND

CALD Victorians may have had fewer opportunities to learn to swim — for reasons of cost, culture, or country-of-origin practice. The drowning data reflects that.

03 · SOCIOECONOMIC STATUS

Lower-income children are less likely to have learned to swim. Swim lessons aren't free; nor is travel to the coast.

04 · GEOGRAPHY

Rural and regional Victorians swim at unpatrolled inland waterways at higher rates than urban residents — because that's the water they have.

05 · ALCOHOL EXPOSURE

Alcohol is a factor in ~20–30% of adult drowning deaths in Victoria. The social settings in which drinking-and-water mix are themselves structural.

06 · COMMUNITY CAPACITY

Local clubs, multilingual programs and trained bystanders aren't distributed evenly. Where they're missing, drowning is more common.

# What a Victorian secondary school could actually do

40% of 2023–24 deaths were CALD community members. Standard curriculum alone is not closing this gap. **In pairs · propose three specific actions** — beyond the standard swim & water safety lessons — that your school could realistically implement.

EXAMPLE 01

**Multilingual beach orientation**

A start-of-summer assembly delivered with EAL/CALD parent groups. Languages chosen from school enrolment data.

EXAMPLE 02

**Subsidised lessons partnership**

Work with LSV / your local aquatic centre on subsidised secondary-aged lessons for students who haven't had access.

EXAMPLE 03

**Peer educator program**

Train senior students from CALD communities to deliver short safety briefings to their own family and community networks.

**Name ONE social determinant a school can address. ONE it cannot.**

Then write a sentence on **why that distinction matters** when you're planning a real prevention strategy.

# 04

## Prevention strategies & international comparison

VCHPEP144

VCHPEP148

INDIVIDUAL + RESEARCH

83%

**Reduction in child drowning in home pools attributable to four-sided isolation pool fencing.**

The strongest evidence base of any single drowning-prevention intervention. **Engineering > education** when the engineering is right.

Hold this number when we compare strategies on the next slides.

# Six categories of prevention — not all created equal

A · ENGINEERING

## Pool fencing, design

4-sided isolation fencing reduces home-pool drowning by ~83%. Strongest evidence of any single intervention.

B · SUPERVISION

## Lifeguards & patrols

All 54 Victorian 2023–24 drowning deaths were at unpatrolled locations. Where patrols are, deaths aren't.

C · EDUCATION

## Swim lessons

Strong evidence for child drowning. For adults, education works best *combined* with behaviour change.

D · LEGISLATION

## Laws & standards

Pool-fencing laws, lifejacket requirements, alcohol restrictions near water. Effective when enforced.

E · TECHNOLOGY

## AEDs, detection, CCTV

Defibrillators at aquatic facilities, drowning-detection systems, throw-rescue equipment at public sites.

F · COMMUNITY

## Programs & campaigns

SwimSafe, Swim & Survive, targeted CALD programs. Strength depends on uptake within high-risk groups.

# Five strategies, ranked against the evidence

| STRATEGY                              | EVIDENCE          | BEST TARGET GROUP                      | KEY LIMITATION                                            |
|---------------------------------------|-------------------|----------------------------------------|-----------------------------------------------------------|
| Pool fencing (4-sided)                | STRONG            | Children under 5 in home pools         | Only addresses backyard-pool drowning                     |
| Mandatory primary-school swim lessons | STRONG            | Children gaining first swim competence | Doesn't reach those who miss school programs              |
| Extending patrol coverage to 7 days   | MODERATE          | Beach-goers in patrolled zones         | Doesn't reach inland waterways at all                     |
| Warning signs at unpatrolled beaches  | LIMITED           | First-time visitors who read signs     | Easily ignored; ineffective with alcohol or peer pressure |
| Targeted CALD community programs      | MODERATE → STRONG | The 40%-over-represented group         | Requires sustained funding & trusted partners             |

**Your call · \$5m budget:** which two strategies would you fund? (Workbook 4.1a — defend with evidence, not intuition.)

# Australia, the UK, the US — what each says first

AUSTRALIA · RLSSA

## "Swim between the flags."

Patrolled-location centric. Beach-flag literacy, SwimSafe rules, Swim & Survive. Inland waterways named as highest-risk environment.

[royallifesaving.com.au](http://royallifesaving.com.au)

UNITED KINGDOM · RLSS UK

## "Float to Live."

Cold-water-shock centric. The primary survival message is a single physiological action — lean back, float, conserve until cold-shock passes.

[rlss.org.uk/education](http://rlss.org.uk/education)

UNITED STATES · RED CROSS

## "Water competency."

Skill-centric framework — swimming, breath control, rescue, and lifejacket use packaged as a competency, not a slogan.

[redcross.org](http://redcross.org) · [take-a-class](#)

**In your workbook:** identify ONE international element *not* currently prominent in Victorian education. Argue whether Victoria should adopt it — citing one primary source URL.

# Which ONE prevention strategy has the strongest evidence? Which has the weakest?

Defend both. One sentence each. **"It feels right"** isn't evidence.

# 05

## Diving, spinal injuries & high-risk teenage behaviour

VCHPEP143

VCHPEP144



INDIVIDUAL + DISCUSSION

# Six numbers that should appear in every prevention message

UNSAFE DIVING — SHARE OF

10%

Of all Australian spinal cord injuries.

UNSAFE DIVING — CAUSE OF

20%

Of quadriplegia cases in Australia.

AQUATIC SPINAL PATIENTS · MALE

95%

A near-total gender skew. **Structural — not coincidental.**

AQUATIC SPINAL · UNDER 30

75%+

Are aged under 30. Highest single age band: **10–14.**

PERMANENT DISABILITY

66%

Of aquatic spinal patients live with permanent disability after the injury.

DROWNING 15–20 YO · POST-COVID

+34%

Increase among 15–20 year olds vs pre-COVID baseline (RLSSA).

# Three locations. One pattern.

CASE 01 · NSW 2025

## Terrigal Wharf

A teenager dives from a wharf into water he believed was deep. Helicopter evacuation. **Hidden shallow zone** under structure.

"IT LOOKED DEEP ENOUGH"

CASE 02 · VICTORIA 2025

## Torquay Front Beach

Teen suffers a suspected spinal injury after a dive at a Victorian surf beach. Airlifted. **Sandbanks shift with every tide.**

SURF BEACHES ARE NOT "SAFE" FOR DIVING

CASE 03 · NSW 2024

## Bawley Beach jetty

A young man is injured diving from a jetty he had used safely before. **New sandbank had formed** beneath the jump zone.

"I'VE DONE IT BEFORE" · FALSE CONFIDENCE

IN PAIRS · 8 MIN · Two factors common to all three. Mechanism — not just behaviour. (Workbook 6.1a.)

# Five factors that compound on rivers, dams and lakes

- 01

**Depth changes constantly**  
Drought, flooding, irrigation releases, sediment movement — even morning to afternoon.
- 02

**Visibility is poor**  
Submerged logs, branches, debris, even shopping trolleys — invisible under turbid water.
- 03

**Cold-water shock**  
Rivers and alpine lakes stay cold even in hot weather. The 1-10-1 rule applies.
- 04

**Alcohol + group dynamics**  
Inland incidents disproportionately involve alcohol, teenage males, and peer-pressured jumps.
- 05

**Remote rescue delay**  
Limited phone signal, no lifeguards, no AED, slower spinal immobilisation — all worsen outcome.

# Six things teenagers believe about diving that are wrong

MYTH

"If others are jumping there, it must be safe."

**FACT** · depth changes daily. Yesterday's safe jump can be today's spinal injury.

MYTH

"Clear water means deep water."

**FACT** · clarity is independent of depth. Dangerously shallow water can be crystal clear.

MYTH

"I checked it before — I know this spot."

**FACT** · Bawley Beach. Familiarity becomes overconfidence becomes injury.

MYTH

"Good swimmers are safe from diving injuries."

**FACT** · spinal injury happens on impact. Swimming ability is irrelevant.

MYTH

"Feet-first is just as risky as diving."

**FACT** · feet-first dramatically reduces cervical spine risk. "Feet first, first time."

MYTH

"The jetty is high — so the water must be deep."

**FACT** · structure height tells you nothing about water depth beneath it.

# Feet first, first time.

Never dive into unknown water. Check depth **every single time**, not just the first time. Previous safety at a location does not guarantee current safety.

**The highest diving-injury incidence is in 10–14 year olds. Explain why — using social *and* environmental factors.**

Then sketch **what a targeted prevention response** would look like. Two sentences each.

# 06

## Assessment task — planning, drafting & peer review



VCHPEP148

INDIVIDUAL + PEER REVIEW

# Three options — all evidence-based, all argued

OPTION A

## Policy Brief

600–800 words to the Victorian Minister for Sport & Recreation. Three recommendations + evidence + measurable outcomes.

Best for: writers who want to argue policy

OPTION B

## Communication Campaign

Target ONE high-risk group from Victorian data. Campaign name, message, visual concept, platform + 100-word justification.

Best for: designers, video-makers, social

OPTION C · ADVANCED ★

## School Audit & Report

Audit your school's water-safety provision. Benchmark against Curriculum 2.0. Priority matrix + recommendations.

Best for: students who want real impact

**All options require:** 3+ credible sources with URLs · 1+ Victorian or global statistic · a named target audience · a specific call to action.

# Six things that separate "Building" from "Extending"

- 01

**Beyond description — to analysis**  
Argues with evidence. Acknowledges counter-arguments. Doesn't just describe what is.
- 02

**Primary sources, cited with URLs**  
LSV, RLSSA, WHO, AIHW — not random blog posts. The reader can verify every claim.
- 03

**Structural — not just individual**  
Names a mechanism. "Masculinity norms" beats "boys take more risks." Distinguishes systemic from personal.
- 04

**Built for a named audience**  
A Minister's brief doesn't read like a TikTok script. The language fits the addressee.
- 05

**Measurable outcomes**  
SMART metrics. "Reduce CALD drowning deaths by 20% in 5 years" beats "raise awareness."
- 06

**Your voice is visible**  
Not a stitched-together summary. Your teacher should feel *you* wrote this and made the calls.

# Six primary sources for the assessment task

## Life Saving Victoria

`lsv.com.au` · `lsv.com.au/research`

Use for · Victorian drowning data, CALD over-representation, SwimSafe rules, patrolled beach locations.

## World Health Organization

`who.int/drowning`

Use for · Global drowning data, prevention guide, social determinants framework.

## International comparators

`rlss.org.uk/education` · `redcross.org/water-safety`

Use for · Float to Live, water competency, international approach comparison.

## Royal Life Saving Australia

`royallifesaving.com.au`

Use for · National data, inland waterway research, post-COVID trends, "Feet first, first time."

## AIHW

`aihw.gov.au`

Use for · Spinal cord injury data, quadriplegia statistics, Australian injury surveillance.

## Victorian Curriculum 2.0

`victoriancurriculum.vcaa.vic.edu.au`

Use for · Option C audit · benchmarking school provision against curriculum codes.

# Six lessons. One thing each.

|                                                                                                                                                                                                        |                                                                                                                                                                                 |                                                                                                                                                                                                           |                                                                                                                                                                                                             |                                                                                                                                                                                         |                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LESSON 01                                                                                                                                                                                              | LESSON 02                                                                                                                                                                       | LESSON 03                                                                                                                                                                                                 | LESSON 04                                                                                                                                                                                                   | LESSON 05                                                                                                                                                                               | LESSON 06                                                                                                                                                                                   |
| <b>The data</b>                                                                                                                                                                                        | <b>The body</b>                                                                                                                                                                 | <b>The system</b>                                                                                                                                                                                         | <b>The evidence</b>                                                                                                                                                                                         | <b>The diving question</b>                                                                                                                                                              | <b>Your argument</b>                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>• 54 Victorian deaths · 2023–24</li><li>• Non-fatals 60% above 10-yr avg</li><li>• ~40% CALD over-representation</li><li>• 0 deaths at patrolled sites</li></ul> | <ul style="list-style-type: none"><li>• 5-stage drowning sequence</li><li>• 4-minute brain window</li><li>• CPR ≈ 30% cardiac output</li><li>• 1-10-1 cold-water rule</li></ul> | <ul style="list-style-type: none"><li>• Individual vs structural causes</li><li>• Six social determinants</li><li>• CALD action a school <i>can</i> take</li><li>• ~80% male skew is structural</li></ul> | <ul style="list-style-type: none"><li>• 4-sided fencing –83% child drowning</li><li>• Six categories of prevention</li><li>• Five strategies, evaluated</li><li>• AU vs UK vs US — three messages</li></ul> | <ul style="list-style-type: none"><li>• 10% / 20% / 95% / 66%</li><li>• Terrigal · Torquay · Bawley</li><li>• Inland = 5 compounding factors</li><li>• Feet first, first time</li></ul> | <ul style="list-style-type: none"><li>• Brief · campaign · audit</li><li>• 3+ sources with URLs</li><li>• Named audience &amp; SMART metric</li><li>• Six rubric criteria · E → A</li></ul> |